TRAINING GUIDE · DIGITAL DATA COLLECTION

KoBoToolbox
A Practical Guide

Design, deploy and manage mobile surveys, from your first form to clean, analysis-ready data. Written for enumerators, M&E officers and research teams.

Form design · XLSForm · KoboCollect · Skip logic and validation · Translations · Data export

Prepared by

Aqil Shah Ayaz

Data Analyst · Monitoring, Evaluation and Learning Expert

Free learning material

aqilshah-researcher.github.io

Contents

Each section has clear learning goals, step-by-step instructions, tips from field use and a short practice exercise.

01 **Getting started** 3
What KoboToolbox is and when to use it

02 **Create your account** 4
Choose a server and sign up

03 **Set up KoboCollect** 6
Install the app, connect it and collect data offline

04 **Build your form** 8
Question types, skip logic and validation

05 **Deploy and manage** 12
Go live, edit safely and share with your team

06 **Add translations** 13
Multilingual forms in the form builder or in Excel

07 **Manage and export data** 14
Review, report, map and download your data

08 **Good practice and quick reference** 16
Survey design rules, checklists and terms

How to use this guide
Work through sections 1 to 5 in order the first time. Sections 6 to 8 are reference material you can return to while you build real forms. Screens change slightly between KoboToolbox releases, so menu names may differ a little from what you see.

INTRODUCTION

01 Getting started

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Explain what KoboToolbox does and why teams use it
- Describe the full data cycle it supports
- Name the main features you will use in this guide

KoboToolbox is a free, open-source platform for collecting data on phones, tablets and computers. It works fully offline, which makes it a standard tool for needs assessments, monitoring visits, household surveys and evaluations in places with weak or no internet.

Design	Deploy	Collect	Review	Export
Form builder or XLSForm	Publish the form	App or web form	Table, map, reports	Excel, CSV, SPSS

Why teams choose it

FREE

No licence, no server

Create an account and start building. Humanitarian and non-profit users can collect large volumes of data at no cost.

OFFLINE

Built for the field

Data is stored on the device and sent to the server when a connection is available.

STANDARD

XLSForm compatible

Design in the browser or in Excel and move between the two at any stage.

SECURE

You control access

Projects are private by default. Share them with colleagues using view, edit or submit permissions.

Main features

Feature	What it does
Form builder	Drag-and-drop editor with a question library for reusing tested questions.
Data collection	Android app (KoboCollect) or web forms that run in any browser, online or offline.
Data management	View, edit and validate submissions; see photos and GPS points on a map.
Reports	Quick charts and tables, with breakdowns by any question such as sex or region.
Export	Download as Excel, CSV, SPSS labels, GeoJSON or KML, ready for analysis.

SETUP

02 Create your account

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Choose the right KoboToolbox server
- Create and verify your account

Step 1. Choose a server

KoboToolbox runs two public servers. Accounts and projects do not move between them, so pick one and keep your whole team on it.

Server	Address	Best for
Global server	`kf.kobotoolbox.org`	Most users, including researchers and NGOs
European Union server	`eu.kobotoolbox.org`	Organizations that need data stored in the EU, including many humanitarian agencies

Check with your organization first
Many organizations already have a team account or a required server for data protection reasons. Ask before you create a personal account for project work.

Step 2. Sign up

1. Open your chosen server address in a web browser.
2. Click **Create an account**.
3. Enter your name, a username, your work email and a strong password. Add your organization, sector and country.
4. Accept the terms and click **Create account**.
5. Open the email from KoboToolbox and click the link to activate your account.
6. Sign in. You will see the **Projects** page, which is empty until you create your first form.

Use a shared team account carefully
For long projects, create forms under one organizational account and share them with each team member. If a staff member leaves, the project and its data stay with the organization.

PRACTICE · 15 MINUTES

Create your account

Work in pairs and help each other through any errors.

- [ ] Choose the server your team will use
- [ ] Create and activate your account
- [ ] Sign in and find the Projects page
- [ ] Write your username somewhere safe: you will need it in the app

DATA COLLECTION

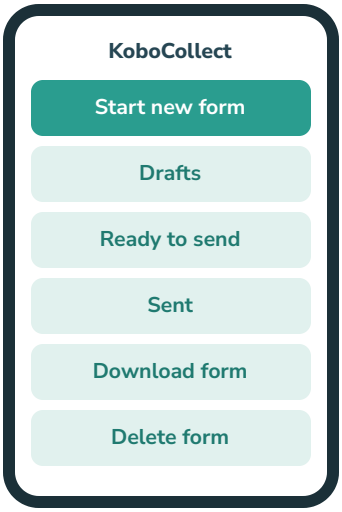
03 Set up KoboCollect

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Install KoboCollect and connect it to your account
- Download a form, collect data offline and send it

Install and connect

1. Install **KoboCollect** from the Google Play Store on an Android phone or tablet.
2. Open the app. Choose **Configure with QR code** if your form page shows one, or **Manually enter project details**.
3. For manual setup, enter the server URL: `https://kf.kobotoolbox.org` or `https://eu.kobotoolbox.org`.
4. Enter your KoboToolbox username and password, then save.



Collect and send data

1. With internet on, tap **Download form** (older versions: *Get Blank Form*), select your form and download it.
2. Tap **Start new form** (older: *Fill Blank Form*) and choose the form.
3. Swipe left to move through the questions. Required questions must be answered before you can continue.
4. On the last screen, tap **Finalize** (older: *Save Form and Exit*, marked as finalized).
5. When you have a connection, open **Ready to send** (older: *Send Finalized Form*), select all and send.

No Android device?
Every deployed form also has a web link (Enketo). It works in any browser on a phone, tablet or laptop, and can also save submissions offline once the page has loaded.

Protect respondents and devices
Use a screen lock on every device, never share passwords between enumerators, and send data daily. Do not leave finalized forms on a phone longer than needed.

PRACTICE · 25 MINUTES

Install and test the app

Use the demo form your trainer has shared.

- [ ] Install KoboCollect and connect it to your account
- [ ] Download the demo form
- [ ] Fill in two test interviews in airplane mode
- [ ] Turn the connection back on and send both
- [ ] Ask your trainer to confirm the submissions arrived

FORM DESIGN

04 Build your form

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Create a project and add questions
- Choose the right question type
- Name your data columns well

Create a project

1. On the Projects page, click **New**.
2. Choose **Build from scratch**, or **Upload an XLSForm** if you designed it in Excel. You can also start from a saved template.
3. Give the project a clear name, a description, sector and country, then click **Create project**.
4. In the form builder, click **+**, type the question text, then pick a question type.

Question types you will use most

Type	Use it for	Type	Use it for
Select One	One answer from a list (yes/no, sex, district)	Text	Short free-text answers
Select Many	Several answers from a list (income sources)	Number / Decimal	Counts, ages, amounts
Date / Time	Visit date, start time	Point (GPS)	Location of a household or site
Photo / Audio / Video	Evidence, verification, consent recording	Note	Instructions shown to the enumerator, no answer
Calculate	Hidden calculations such as totals or scores	Rating / Ranking	Scales and priority lists

Question settings to set every time

Data column name

A short, unique name in English with no spaces, such as `hh_size` or `water_source`. This becomes the column header in Excel, so keep it readable.

Mandatory response

Set to **Yes** for questions that must be answered. Use it carefully: forcing answers to sensitive questions can push enumerators to guess.

Hint

Short guidance for the enumerator, for example "Include children under 5".

Default value

Useful for fixed values, such as the province for a team that only works in one area.

Name columns before you collect
Changing data column names after collection has started splits your data into old and new columns. Agree on a naming convention with the analyst first.

FORM DESIGN

04 Skip logic and validation

Skip logic: show questions only when they apply

Skip logic hides a question unless a condition is met. You always set it on the question that should sometimes be hidden, not on the earlier question.

1. Open the settings of the question to show or hide, then choose **Skip logic**.
2. Click **Add a condition**, pick the earlier question, then the operator and the value. Example: show *Which school does the child attend?* only if *Is the child in school?* = Yes.
3. With two or more conditions, choose whether the question needs **all** of them or **any** of them.

Validation criteria: stop impossible answers

Validation rejects answers outside a sensible range and shows a message to the enumerator, for example an age between 0 and 120, or a household size of at least 1.

The same rules in XLSForm

For larger forms, many teams design in Excel using the XLSForm standard and upload the file. These three columns do most of the work:

type	name	label	relevant	constraint	required
`select_one yes_no`	`in_school`	`Is the child in school?`			`yes`
`text`	`school_name`	`Which school?`	`${in_school} = 'yes'`		`yes`
`integer`	`age`	`Age of respondent`		`. >= 18 and . <= 99`	`yes`
`select_multiple income`	`income_src`	`Sources of income`			`no`

Survey sheet. Options for `yes_no` and `income` are listed in a separate **choices** sheet with the columns list_name, name and label.

Test every branch
Before deployment, preview the form and walk through each path: yes and no answers, minimum and maximum values. Most data errors come from logic that was never tested.

PRACTICE · 40 MINUTES

Transcribe a short questionnaire

Use the 10-question paper form provided by your trainer.

- [ ] Create a new project and enter all ten questions
- [ ] Give every question a clear data column name
- [ ] Add one skip logic rule and one validation rule
- [ ] Preview the form and test both paths of the skip logic

GO LIVE

05 Deploy and manage

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Deploy a form so data collection can start
- Edit a live form safely
- Share a project with your team

Deploy

Open the project and click **Deploy**. The form is now live: it can be downloaded in KoboCollect and opened through its web link. After deployment the button changes to **Redeploy**.

Project actions

Action	What it does and when to use it
Edit	Change questions in the form builder. Changes are saved as a draft until you redeploy.
Preview	Test the form in the browser exactly as respondents will see it.
Redeploy	Publish your edits. Enumerators must download the form again to get the new version.
Replace form	Upload a new XLSForm to replace the current design while keeping the project.
Download XLS	Save the form as an XLSForm, as a backup or to edit translations in Excel.
Share	Add colleagues with permission to view the form, view data, edit, or submit data.
Clone	Copy the form to reuse it as a template for the next survey round.
Manage translations	Add and edit other languages for the form (see section 6).

Editing a form during data collection
Small wording fixes are safe. Do not delete questions or change data column names or choice values after collection has started: existing data will no longer match. If you must change the design, agree on it with the analyst and note the date of the change.

Version control
Download the XLSForm each time you redeploy and save it with the date in the file name, for example `baseline_survey_2026-03-14.xlsx`.

LANGUAGES

06 Add translations

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Set a default language
- Add translations in the form builder
- Add many translations quickly in Excel

Option A. In the form builder

1. Open the project, click **Manage translations**.
2. Set the **default language**: the language the form was first written in. Enter its name and code, for example English with code `en`.
3. Click **Add language** and enter the next language, for example Dari (`prs`) or Pashto (`ps`).
4. Click the **Update translations** icon next to the new language and type each translated question and choice.
5. Redeploy the form. Enumerators choose the language inside the app.

Option B. In Excel (fastest for long forms)

A translator can work in Excel without access to your KoboToolbox account.

1. Click **Download XLS** to export the form.
2. In the **survey** sheet, rename the `label` column to include the language, for example `label::English (en)`.
3. Insert a new column next to it for each language, for example `label::Dari (prs)`, and fill in the translations.
4. Do the same for `hint` and constraint message columns if your form uses them, and for `label` in the **choices** sheet.
5. Back in KoboToolbox, choose **Replace form**, upload the file and redeploy.

name	label::English (en)	label::Dari (prs)	label::Pashto (ps)
hh_size	How many people live in this household?	[Dari text]	[Pashto text]

Back-translate key questions
Ask a second person to translate the most important questions back into English. Differences in meaning show up quickly and are much cheaper to fix before fieldwork than after.

RESULTS

07 Manage and export data

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Monitor submissions while fieldwork runs
- Use reports and the map for quick checks
- Download data in the right format for analysis

Summary

The project summary shows the form details, who has access and a chart of submissions over time. Check it daily during fieldwork to spot teams that are behind or not sending data.

Data views

TABLE

Review each record

Filter, sort, edit or mark submissions as approved, on hold or not approved. Use this for daily quality checks.

REPORTS

Quick charts

Automatic charts and tables for every question. Use **Group by** to break results down by sex, region or any other question, and save custom reports for the indicators you track.

GALLERY

Photos and media

See all photos collected, for example to verify site conditions or distributions.

MAP

GPS points

See where interviews took place and color the points by an answer, for example by sex or by visit outcome.

Download your data

1. Open **Data** and then **Downloads**.
2. Choose the format: **XLS** or **CSV** for most analysis, **SPSS labels** to label variables in SPSS, **GeoJSON** or **KML** for maps.
3. For multilingual forms, choose the language for labels, or XML values for the raw codes.
4. Click **Export**, wait for the file to be ready and download it.

Handle personal data with care

Downloads may contain names, phone numbers and locations. Store them in a secure, access-controlled folder, remove identifiers before sharing, and follow your organization's data protection policy.

REFERENCE

08 Good practice and quick reference

Golden rules of survey design

1. **Start from the indicators.** Every question should feed an indicator or a clear analysis need.
2. **Keep it short.** Long interviews tire respondents and lower data quality.
3. **One idea per question.** Avoid "Do you have water and electricity?".
4. **Use plain words** that respondents use every day.
5. **Add "Other, specify" and "Don't know"** where they are real options.
6. **Order from easy to sensitive.** Put consent first and sensitive topics later.
7. **Use logic, not instructions,** to skip questions that do not apply.
8. **Validate numbers** so impossible values cannot be entered.
9. **Pilot the form** with real respondents before training the full team.
10. **Freeze the design** before collection starts, and document any later change.

BEFORE DEPLOYMENT

- [ ] Consent question at the start
- [ ] All column names agreed and unique
- [ ] Skip logic and validation tested
- [ ] Translations checked
- [ ] Form piloted and fixed
- [ ] XLSForm backup saved

DURING FIELDWORK

- [ ] Submissions checked daily
- [ ] Outliers and gaps followed up
- [ ] GPS points reviewed on the map
- [ ] Devices synced and phones locked
- [ ] Data exported and backed up weekly
- [ ] Issues logged with dates

Key terms

Deploy
Publish a form so it can be used to collect data.

XLSForm
A standard for writing forms in Excel, used by KoboToolbox and ODK.

Data column name
The variable name that becomes a column in your export.

Skip logic (relevant)
A rule that shows a question only when a condition is true.

Validation (constraint)
A rule that rejects answers outside an allowed range.

Enketo
The web form engine that runs your form in a browser.